THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET B

Chapters 13 & 14: **Two Variables, One Line, Math of Space: Surface Area and Volume** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The slope of the line through the points (-3, -2) and (1, 4) is: 1

(a) -3/2 **(b)** 3/2 **(c)** 2/3 **(d)** 5/2

2. The y-intercept of the line $3x - 4y = 12$ is: 1

(a) -3 **(b)** -4 **(c)** 4 **(d)** 3

3. The pair of equations $2x - 3y = 5$ and $4x - 6y = 10$ has: 1

(a) no solution **(b)** exactly two solutions

(c) infinitely many solutions **(d)** a unique solution

4. For which value of k does the pair $kx + 2y = 4$ and $3x + 6y = 12$ have infinitely many solutions? 1

(a) 3 **(b)** 2 **(c)** 1 **(d)** 6

5. 2 adult tickets and 3 child tickets cost ₹1050, while 3 adult tickets and 2 child tickets cost ₹1200. The price of one adult ticket is: 1

(a) ₹300 **(b)** ₹450 **(c)** ₹150 **(d)** ₹250

6. Taking $\pi = \frac{22}{7}$, the surface area of a sphere of radius 7 cm is: 1

(a) 308 cm^2 **(b)** 616 cm^2 **(c)** 1232 cm^2 **(d)** 154 cm^2

7. Taking $\pi = \frac{22}{7}$, the curved surface area of a hemisphere of radius 7 cm is: 1

(a) 462 cm^2 **(b)** 308 cm^2 **(c)** 616 cm^2 **(d)** 154 cm^2

8. A pyramid has a base of area 48 cm^2 and a height of 9 cm. Its volume is: 1

(a) 144 cm^3 **(b)** 432 cm^3 **(c)** 216 cm^3 **(d)** 72 cm^3

9. The surface area of a sphere of radius r equals the curved surface area of a cylinder that tightly fits over it. The height of this cylinder is: 1

(a) r **(b)** $2r$ **(c)** πr **(d)** $4r$

10. **Assertion (A):** The surface area of a sphere of radius r equals the curved surface area of a cylinder of radius r and height r. 1

Reason (R): The curved surface area of a cylinder is $2\pi rh$.

(a) Both A and R are true and R is the correct explanation of A

(b) Both A and R are true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Without solving, decide whether $3x + 2y = 5$ and $6x + 4y = 11$ have a solution. Give a reason. 2

12. A ramp rises 3 m over a horizontal run of 12 m. Find its slope and state what it means. 2

13. The radii of two spheres are in the ratio $1 : 2$. Find the ratio of their surface areas. 2

14. If the edge of a cube is doubled, by what factor do its surface area and volume change? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write $y = \frac{2}{3}x - 4$ in the form $ax + by + c = 0$ with integer coefficients and verify that $(6, 0)$ and $(3, -2)$ lie on it. 3

16. The base circumference of a hemispherical dome is 66 m. Find the cost of painting its outer curved surface at ₹40 per m^2 ($\pi = \frac{22}{7}$). 3

17. A wheelchair ramp must have a slope of 1 : 12 (rise : run). If the height of a platform is 30 cm, what is the minimum horizontal length of the ramp? Express the answer in metres. 3

OR

If $(2, -1)$ is a common solution of $ax + 3y = 1$ and $4x + by = 5$, find a and b. 3

18. The edge of a cube is increased by 50%. Find the percentage increase in 3
(i) its total surface area,
(ii) its volume.

OR

A cylindrical glass has radius 3 cm and contains water to a height of 10 cm. How many marbles of radius 1 cm must be dropped in so that the water level rises to 14 cm? (Marbles sink fully.) 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A shop sells two combo packs. Pack 1 has 2 pens and 3 notebooks for ₹80; Pack 2 has 3 pens and 2 notebooks for ₹70. 5
(i) Form the pair of equations in the pen price x and notebook price y.
(ii) Solve it.
(iii) What is the cost of 1 pen and 1 notebook together?
(iv) What does a unique solution mean here?

OR

A boat goes 20 km downstream in 2 hours and 12 km upstream in 2 hours. Find the speed of the boat in still water and the speed of the stream. 5

20. A solid metal cylinder of radius 3 cm and height 8 cm is melted and recast into small solid spheres of radius 1.5 cm. 5
(i) Find the volume of the cylinder.
(ii) Find the volume of one sphere.
(iii) How many spheres are formed?
(iv) Find the total surface area of all the spheres. Leave answers in terms of π.

OR

The radius of a cylinder is increased by 20% while its height is unchanged. Find the percentage change in 5
(i) the curved surface area,
(ii) the volume, and
(iii) the total surface area when $h = r$.

END OF QUESTION PAPER — **Total: 40 Marks**